

CHASE MEMMER

ChaseMemmer@gmail.com ◇ 727-410-1943 ◇ Tampa, FL

EDUCATION

University of Alabama

M.S. in Mechanical Engineering - 2025

B.S. in Mechanical Engineering (Cum Laude) - 2022

B.S. in Electrical Engineering (Cum Laude) - 2022

**Multidisciplinary R&D hardware engineer
building integrated electrical and mechanical
systems.**

WORK EXPERIENCE

AEVEX Aerospace

Electrical Engineer

Tampa, FL

July 2024 – Present

- Subsystem lead or sole electrical engineer on 6 projects; owned architecture/requirements from concept to production
- Owned design validation processes, including design reviews and FMEA; mentored junior engineers and interns
- Contributed to mechanical and systems design to support integrated product development
- Designed system, control loop and electrical architecture of multi-drone launcher
- Led 5-engineer team developing drone ground control subsystem with charging, communication, and launch systems
- Designed multi-frequency, long range (>1000 mi), jam-resistant radio for telemetry balloon
- Redesigned acquired Group 3 drone, cutting cost and enabling scalable production
- Led team to introduce DFx into production workflow

DEKA Research and Development

Electrical Engineer (Acting Electrical Lead)

Manchester, NH

January 2023 – July 2024

- Sole electrical engineer on 35-person multidisciplinary team; owned all power and signal design decisions
- Defined electrical architecture and derived requirements from system-level specifications
- Designed PLC/RIO panels, wiring schematics, and cable harnesses (2D/3D) in SolidWorks Electrical
- Developed and validated circuit designs using LTSpice simulation and hardware testing
- Designed and laid out 4+ layer PCBAs for power distribution, safety, and mixed-signal systems (Altium)
- Led electrical verification and compliance efforts to meet UL, ISO, NFPA, and CFR standards

Quincy Air Compressor

R&D Engineering Co-Op

Bay Minette, AL

May 2019 – May 2020

- Designed test procedures and analyzed performance data using LabVIEW and Excel VBA
- Managed third party contractors and implemented compliance with ISO, ASME, DOE, and OSHA standards

PROJECTS

Electromyography-Controlled Prosthetic Arm – Master's Research

Matlab, Research

Tuscaloosa, AL

January 2024 – June 2025

- Researched EMG-based control methods for biomimetic prosthetic systems

Battery Management System for ABEX Satellite

Team Lead, Embedded Hardware Engineer, Embedded Systems

Tuscaloosa, AL

January 2022 – January 2023

- Designed energy management system (EMS) for open source satellite, including power optimization, storage, and distribution
- Led small team; owned hardware design, part selection, budgeting, and documentation

NASA's Robotic Mining Competition

Mechanical Team Lead, Mechanical Engineer

Tuscaloosa, AL

September 2018 – May 2020

- Led mechanical redesign that doubled production rate and won national championship
- Fully modelled and simulated 1000+ part system in Solidworks
- Served as technical lead; presented design at NASA consortium in Washington, D.C.

SPECIALIZED SKILLS

Domains

Defense, Robotics, Medical

Design

System Architecture, Mixed-signals PCBA, Controls, Sensor Integration, PLC

Tools

Altium, KiCAD, LTSpice, Solidworks, Solidworks Electrical

Programming

MATLAB, C, Arduino, Python

Hardware

PCB Bring-up, debugging, instrumentation, soldering/rework, wiring, prototyping